INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, Second Semester Examination, March – 2026

Subject Code: BCACC202

Subject Name: Object Oriented Programming

Full Marks: 30 | Set No.3 | Duration: 1 Hour

Part – A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1.	Discuss the anomaly that occurs in multiple inheritance.	2
2.	What are virtual base classes?	2
3.	What is the purpose of throwing an exception in C++?	2
4.	What are the benefits of using pointers?	2
5.	What is hierarchical inheritance? Give an example.	2
6.	How does late binding support polymorphism in C	2

Part – B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7.	How virtual base class solve the anomalies in hybrid inheritance?	5
8.	Explain how virtual functions support runtime polymorphism in C++.	5
9.	What is operator overloading? Why is it necessary to overload an operator?	5

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Part – C

Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Implement a C++ program demonstrating multiple inheritance. **10**

11. Define a class Complex to represent a complex number with real and imaginary parts. **10**

a) Overload the + operator to add two complex numbers.
b) Overload the == operator to compare two complex numbers.
c) Overload the << operator to display a complex number in the format: a + bi.
d) Write a main () function to demonstrate all the overloaded operators.

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